VALIDATION OF PROCESSES, SERVICES OR PRODUCTS

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01	Initial edition	22/02/2024	N/A
02	Inclusion of a protocol for validating software development tools, updating Spika departments	June 14, 2024	PNT-018.01
03	Change in Spika's departments	02/06/2025	PNT-018.02

Content

1. Objective 4
2. Scope 4
3. Responsibilities and Departments affected 4
4. Definitions and Acronyms 5
5. Procedure 5
5.1. Generalities 5
5.2. Types of validation 6
5.3. Methodology 6
5.3.1. Software 6
5.3.2. Development Tool 7
5.3.3. Production processes 8
5.4. Documentation 9
5.5. Re-validation 9
6. Distribution of Procedure 10
7. Annexes 10
8. Related documentation 10
9. Formation 11

1. Aim

The objective of this procedure is to establish the general validation policy applied by Spika Tech, hereinafter SPIKA, for the products it supplies.

2. Scope

This procedure covers everything from the definition of user requirements to product validation. This document defines the general scope of validation and the activities involved, although the strategy and planning of specific activities will be included in the protocols, and their execution and results will be documented in the corresponding reports.

3. Responsibilities and Departments Affected

It is the responsibility of all SPIKA employees to collaborate and manage product validation activities.

The Technical Manager and the Research and Development staff are responsible for supervising the execution and compliance of this procedure.

The affected departments are:

Department	Apply	
	Yeah	No
Address	☒	☐
Environment	☒	☐
Quality Assurance	☒	☐
Systems and Cybersecurity	☒	☐
Research and Development	☒	☐
Marketing and Sales	☒	☐

Management	☒	☐

4. Definitions and Acronyms

- Acceptance criteria: Product specifications and acceptance/rejection criteria, such as the acceptable quality level, with an associated sampling plan, that will be necessary to make the decision to accept or reject a product.
- Hardware and Software Installation Qualifications: Documents the installation, configuration, and qualification instructions for hardware and software, including middleware, to demonstrate that all of these items can perform the functions required by the application software.
- Operational Qualification (OQ): establishes, by objective evidence, that the equipment or instrument is capable of operating repeatedly within the determined limits and meeting the specifications.
- Performance Qualification (PQ): establishes, based on scientific evidence, that the equipment or instrument is capable of delivering effective and reproducible results according to the approved process and product specifications.
- Validation: According to the FDA, it is "the establishment of documented evidence that provides a high degree of assurance that a specific process will consistently produce a product that meets its predetermined specifications and quality attributes." According to European standards, validation is "the obtaining of evidence, in accordance with Good Manufacturing Practices, that any procedure, process, equipment, activity, or system actually produces the intended result." Validation refers to processes, systems, and methods.

5. Procedure

5.1. Generalities

Validation is the collection and evaluation of documentation from the design phase of a process through commercial production, establishing scientific evidence that a process is capable of ensuring product quality.

An effective validation process contributes significantly to ensuring product quality.

An important point to consider in the validation exercise will be the changes. Each relevant change within the product must be reviewed and evaluated to determine whether or not it requires revalidation, as well as periodic quality reviews.

For each validation exercise, a validation protocol must be duly reviewed and approved before its execution, and it must be performed by trained personnel. The trained personnel will collect all data, as well as any incidents or nonconformities detected, to issue a final judgment and declare the validation exercise as satisfactory or not in the corresponding report. Where applicable, after detecting nonconformities, corrective actions will be initiated as necessary, and the compliance status of the validation exercise will be assessed.

All generated documentation will be archived in a secure location in the Quality Archive.

5.2. Types of validation

The following types of validation to be performed at SPIKA are listed below; however, any exercises may be evaluated as appropriate to ensure product quality:

- Software validation
- Development tool validation
- Validation of production processes

5.3. Methodology

Newly created products, change controls, or even quality reviews or risk analyses will give rise to validations.

5.3.1. Software

The minimum content that the Computerized Systems Validation Protocol must present is described below:

- Aim.
- Responsibilities.
- Scope.
- Definitions.
- Staff training.
- Study of critical stages and phases

- Methodology.
- Related documentation.
- Annexes.

The software in question must be studied using Risk Analysis and at least the following verifications must be carried out:

- Functional specifications
- Risk Analysis
- Design qualification: Since SPIKA products are developed internally, it must be established that functional specifications correspond to operational requirements and that expectations have been met.
- Installation Qualification (IQ): The installation path and the manner in which the software in question is installed must be qualified.
- Operational Qualification (OQ): where it must be demonstrated that the planned functionalities have been implemented and that they work.
- Performance Qualification (PQ): where the expected results must be verified through actual executions of the application.

5.3.2. Development tool

The minimum content that the software development tool validation protocol must present is described below:

- Introduction
- Scope
- Definitions
- Validation requirements
- Validation plan
- Validation procedures
- Results and conclusions

This framework will ensure complete and rigorous validation of the software development tool, ensuring that it meets the standards required for effective and secure use in its intended context.

5.3.3. Production processes

Process validation requires that the process is carried out using qualified facilities, systems, and equipment, that the corresponding methods have been validated, that work procedures are in place, and that personnel have been trained in these procedures.

The validation strategy is reflected in the specific validation protocol. Acceptance criteria must be defined in advance, documented, and justified in the validation protocols, based on the results of the AR prepared in each case.

Process validation must be completed prior to product distribution and sale.

A corresponding validation protocol will be established for each process, defining the type of validation, the number of batches to be validated, the number and quantity of samples to be collected, and the analyses to be performed. If statistically, a result of one or two batches to be validated is obtained, the validation must be performed considering three batches.

In the concurrent and prospective validation approach, process validation is typically carried out with a number of consecutive batches established based on the results of the AR. Deviations from this system may be permitted under justified circumstances.

Likewise, a validation may concern only the manufacturing or packaging phase because the modification requiring validation only affects one of the phases. This will be clearly defined in the corresponding validation protocol.

They all have a common format in which the following points are defined:

- Aim.
- Responsibilities.
- Scope.
- Teams.
- Definitions.
- Staff training.

- Study of critical stages and phases (Risk Analysis).
- Process(es) for Validation: Manufacturing and Packaging, Manufacturing or Packaging.
  - Development of Validation.
  - Environmental conditions during process validation.
  - Waste Management.
- Related documentation.
- Annexes.

5.4. Documentation management

The protocols will be coded as follows:

- PV-XXX/YY.ww , where XXX is a consecutive number, YY the year of issue and ww the version (always starting with 01).

Validation reports will be coded as follows:

- IV-XXX/YY.ww , where XXX is a consecutive number, YY the year of issue and ww the version (always starting with 01).

For every protocol, review by the quality unit and approval by at least the responsible technician will be required. Then, as many figures as the validation exercise requires will be involved.

5.5. Re-validation

Validation is not a one-time activity, but rather a continuous one. To achieve this, a system must be defined that detects unplanned deviations from the designed process, thus enabling control to be maintained or improved and process and product variability to be reduced.

Continuous process verification through monitoring and trend analysis will determine whether process changes are necessary. All of these efforts should identify process variability and identify potential improvements.

To ensure that processes and related procedures are robust and that product quality is consistent, periodic process revalidation or continuous process verification must be performed. This ongoing process verification must be applied throughout the product lifecycle to support the validated status of the product. The Organization's Change Control tool and the conclusions of the Management Review Report will be used as a basis, and an annual evaluation will be made of whether or not revalidation is appropriate.

6. Distribution of the Procedure

Staff members in the departments listed in Section 3 (Responsibilities and Affected Departments) must be provided with a copy of this procedure. In addition, controlled copy No. 1 will be submitted to the Archives for paper archiving.

Controlled copies to be issued:

Controlled Copy No.	Departments
1	Archive

7. Annexes

NO. / REV.	QUALIFICATION
N/A	N/A

8. Related documentation

Code	QUALIFICATION
PNT-005	Change Control Management
PNT-004	Non-Conformity Management

9. Training

This version requires that the affected departments indicated in section 3 receive the training indicated below:

Mark with an X	Type of training
☒	Theoretical (reading and understanding of the procedure)
☐	Theoretical – Practical (If you select this option, contact the Responsible Technician)